FIFA WORLD CUP

Introduction

You have been assigned the task of designing the FIFA network. For simplicity, you have been asked to design the hotels only for 6 countries. The countries are France, Spain, Germany, Argentina, Brazil, and Portugal. The total number of staff/hosts needed for each hotel is given below.

France -> 200 Hosts
Spain -> 100 Hosts
Germany -> 150 Hosts
Argentina -> 200 Hosts
Brazil -> 250 Hosts
Portugal -> 300 Hosts

Requirements

Topology Design & Server Setup

- Every country's hotel is represented by a router.
- Brazil's hotel is only connected to Argentina's hotel.
- Argentina, Spain and France's hotels are connected to one another via a switch
- There is a loop among the European countries' hotels. France is directly connected to Portugal, Portugal is connected to Spain, Spain is connected to Germany, and then Germany is connected to France.
- All the hotels have their own Email Servers set up. You have to configure the emails for communication. The hostnames are like mail.argentina.com, mail.brazil.com and so on.

- Only Argentina (www.argentina.fifa.com) and France (www.france.fifa.com) have dedicated web servers. You have to configure the web servers so that all devices can visit them.
- Only Argentina's hotel has the central DNS server. You have to configure the DNS server so that all devices can use it.
- France's router is working as the DHCP server for all the hotels except for Argentina & Brazil. Argentina & Brazil have their dedicated DHCP servers. You have to configure this.
- Each hotel has a printer, and you can show only two PCs to represent all the devices of that particular hotel.

Addressing

- Select the student ID of the first member of your group. Now take the last 4 digits of that ID and divide them into two parts containing two digits each. These two parts are the first two octets of the network address of the FIFA network. The third and fourth octets are 0, and the prefix mask is 16. Suppose the ID is 20201002. Then the last four digits are 1002. Now dividing it into two parts gives 10 and 02. So the Network address is 10.2.0.0/16.
- Subnet this network using VLSM to provide a network address to each hotel's network.
- All the PCs of the hotels will get IP addresses dynamically, while the servers, routers and printers will be configured manually.

Routing

- France, Spain, and Germany's hotels will share their routing table with one another using RIP protocol.
- France's hotel is aware of all possible networks, either dynamically or statically.
- Brazil's hotel can reach other networks only through Argentina, and so a default path needs to be set up.
- Argentina and Portugal know about other networks using static routing.

- If the path between Portugal and France is down, then Portugal sends the packets through another path using recursive routing.

You are allowed to make any valid and necessary assumptions while designing the network infrastructure.

Deliverables

- The network mentioned above should be implemented in Cisco Packet Tracer, with the necessary devices and full configuration.
- After completion, you should be able to test the conditions imposed, and all the devices should be able to ping one another.
- You will have to submit the following:
    - Work Distribution among the group members [Who did which part]
    - The pkt/pka file
    - Picture of the Network topology diagram with proper labels [You have to show the network addresses using notes for each network]
    - A PDF containing
        - VLSM tree
        - IP address table
        - The configuration commands of all the routers you have implemented.